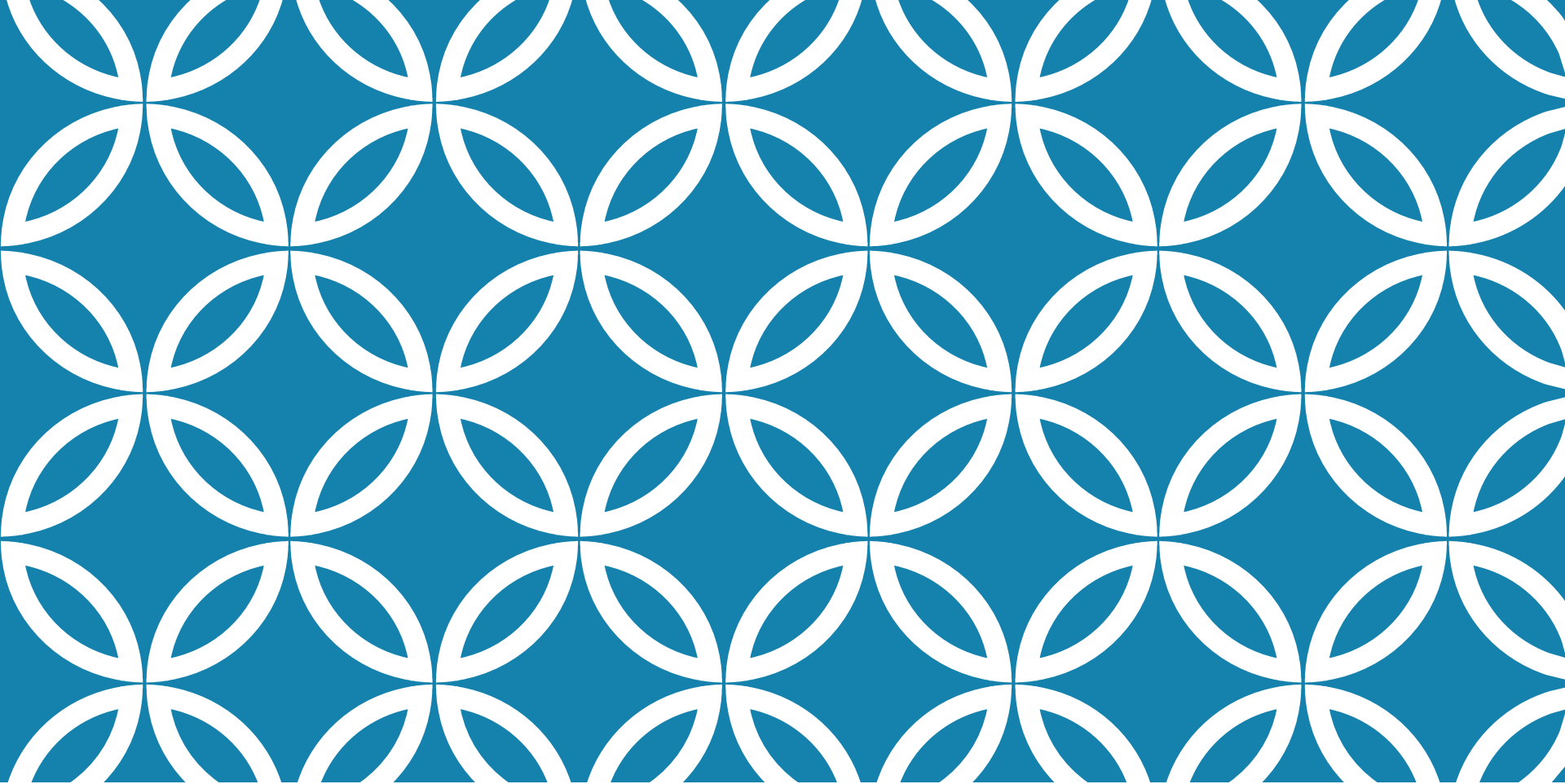


IT SOCIETY

INTRODUCTION

DATABASES

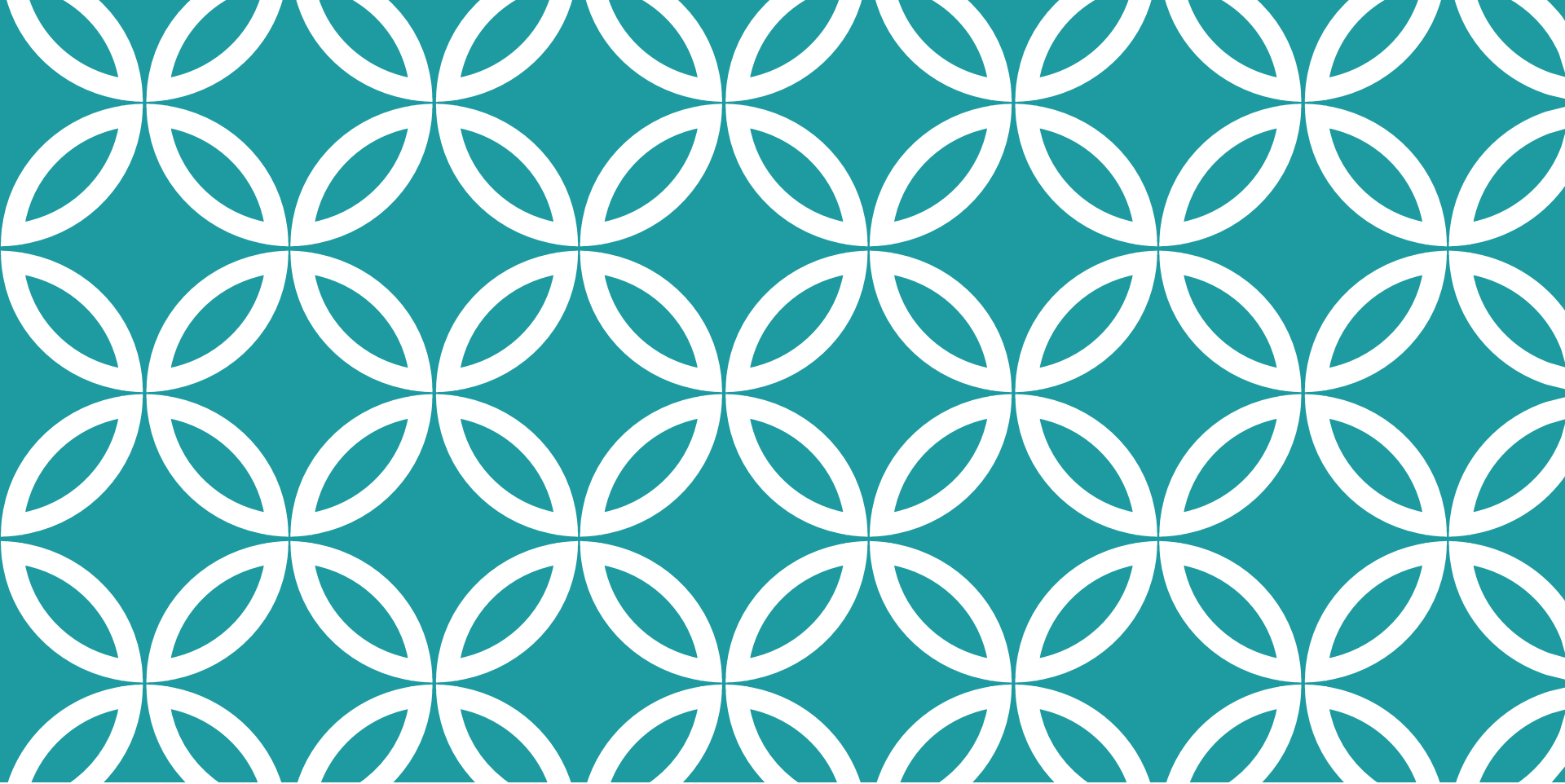
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RELATIONAL DATABASES





TRANSACTION CONTROL LANGUAGE

**TRANSACTION,
COMMIT,
ROLLBACK &
SAVEPOINT**

TRANSACTION CONTROL LANGUAGE (TCL)

- **TCL (Transaction Control Language)** in SQL is used to manage database transactions and maintain data integrity.
- These include;
 1. TRANSACTION
 2. COMMIT
 3. ROLLBACK
 4. SAVEPOINT

7. TRANSACTION

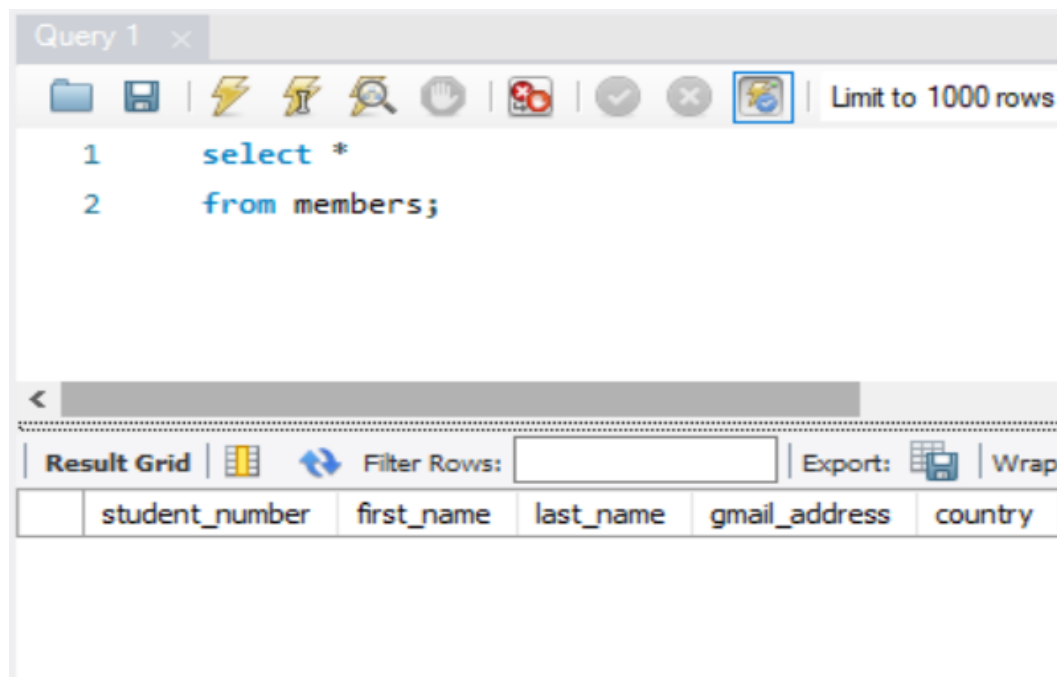
- A **transaction** in a database is a sequence of one or more operations performed on its data.

COMMAND	PURPOSE
START TRANSACTION	Begins a new transaction block

- A transaction can either end with a **COMMIT** or **ROLLBACK** command.

7. TRANSACTION (CONTINUES...)

- Recall that we table has no records –

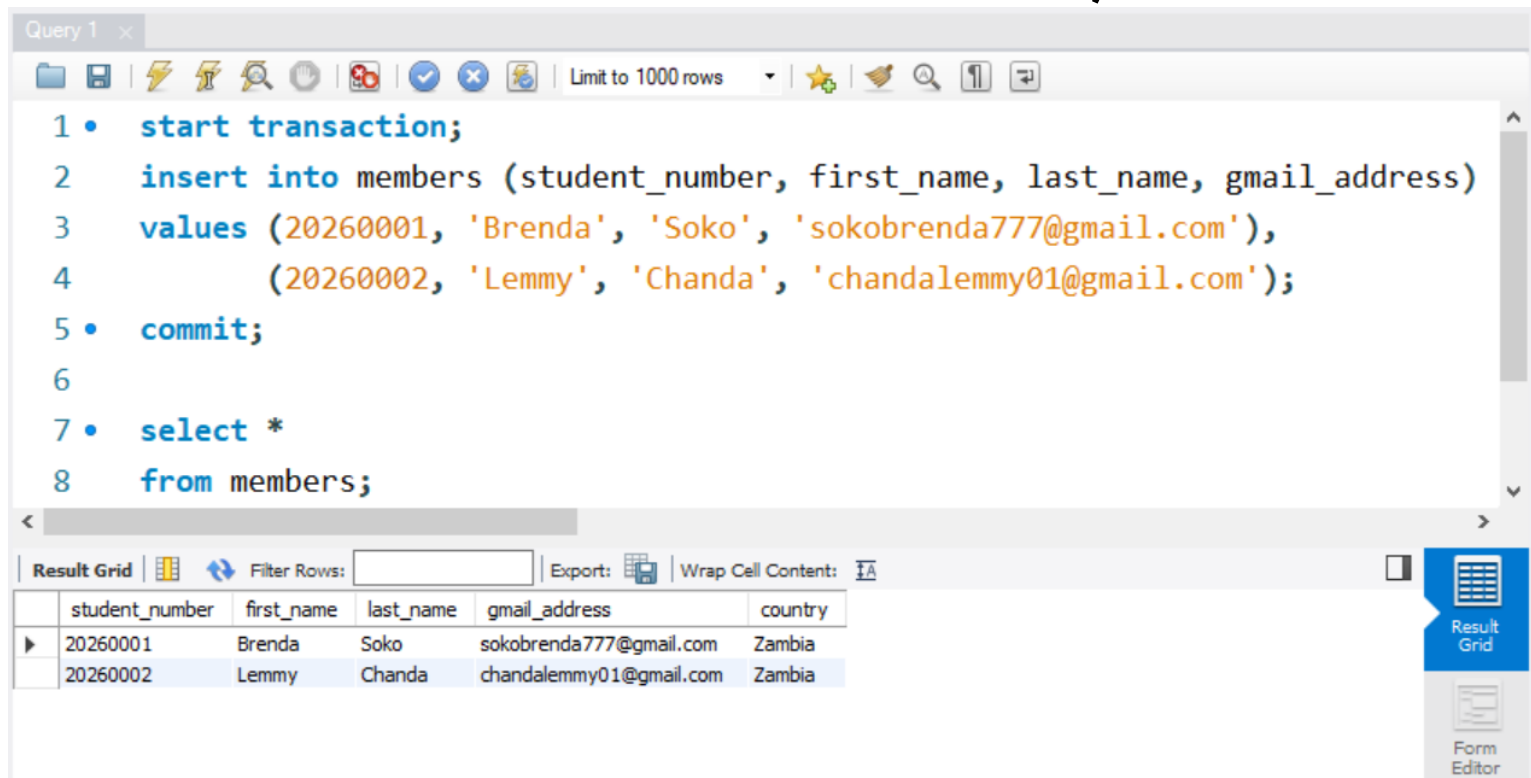


8. COMMIT

COMMAND	PURPOSE
COMMIT	Permanently saves all changes made in the current transaction.

8. COMMIT (CONTINUES...)

- Let's insert some records in our table;



The screenshot shows a database query editor window titled "Query 1". The SQL code is as follows:

```
1 • start transaction;
2 insert into members (student_number, first_name, last_name, gmail_address)
3 values (20260001, 'Brenda', 'Soko', 'sokobrenda777@gmail.com'),
4         (20260002, 'Lemmy', 'Chanda', 'chandalemmmy01@gmail.com');
5 • commit;
6
7 • select *
8 from members;
```

Below the code, the "Result Grid" is displayed, showing the results of the query. The grid has five columns: student_number, first_name, last_name, gmail_address, and country. There are two rows of data:

student_number	first_name	last_name	gmail_address	country
20260001	Brenda	Soko	sokobrenda777@gmail.com	Zambia
20260002	Lemmy	Chanda	chandalemmmy01@gmail.com	Zambia

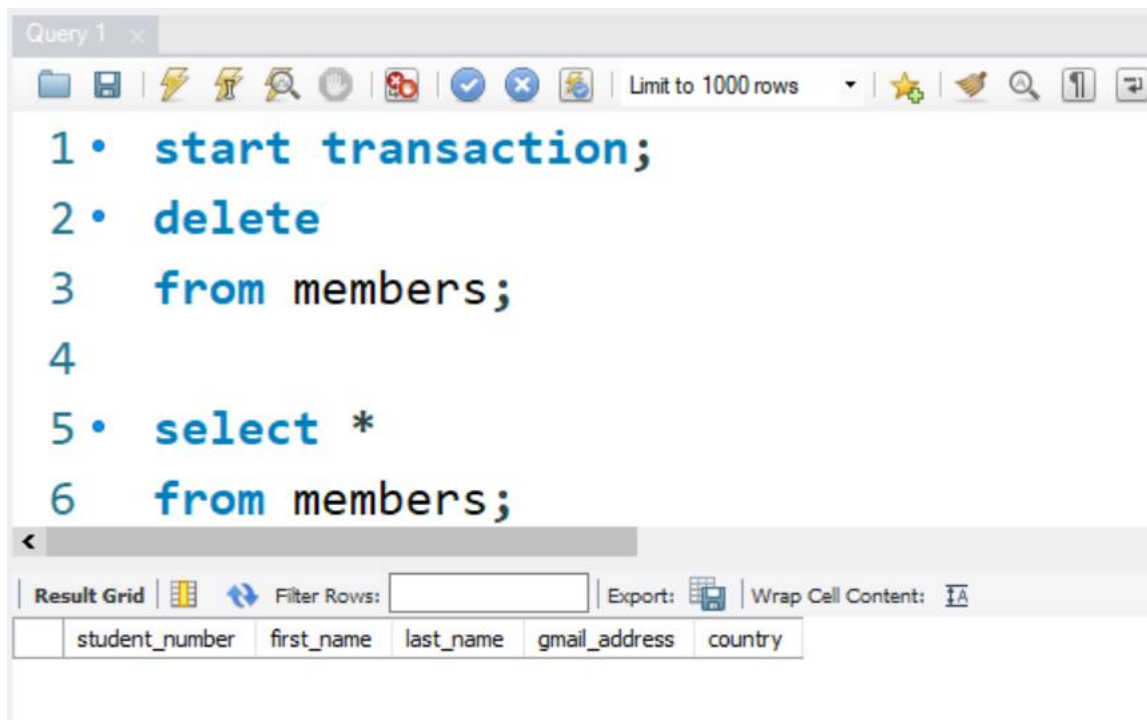
The interface also includes a toolbar with various icons, a "Limit to 1000 rows" dropdown, and a "Filter Rows" input field. The "Result Grid" button is highlighted in blue on the right side of the interface.

9. ROLLBACK

COMMAND	PURPOSE
ROLLBACK	Undoes all changes made in the current transaction

9. ROLLBACK (CONTINUES...)

- Now suppose we delete all records from our table



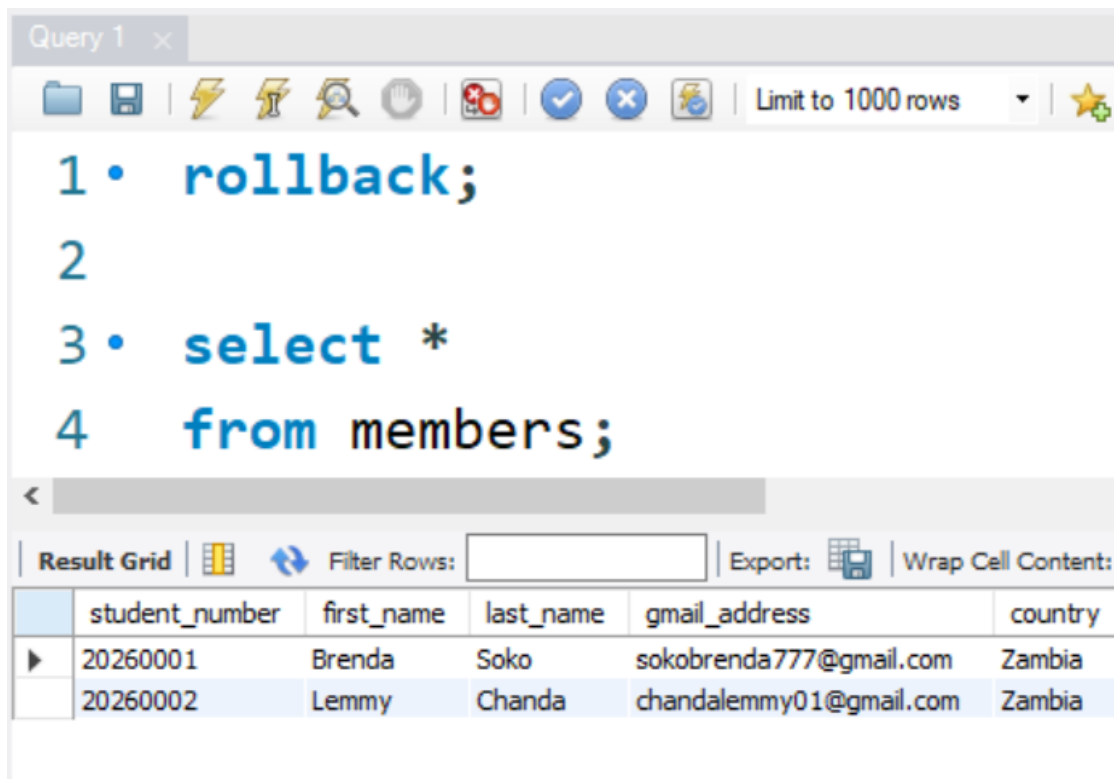
The screenshot shows a database query editor window titled "Query 1". The toolbar includes icons for file operations, execution, and settings, along with a "Limit to 1000 rows" dropdown. The SQL script is as follows:

```
1 • start transaction;  
2 • delete  
3   from members;  
4  
5 • select *  
6   from members;
```

Below the script, the "Result Grid" section is visible, showing a table with the following columns: student_number, first_name, last_name, gmail_address, and country.

9. ROLLBACK (CONTINUES...)

- Let's restore our records to the previous SAVEPOINT



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

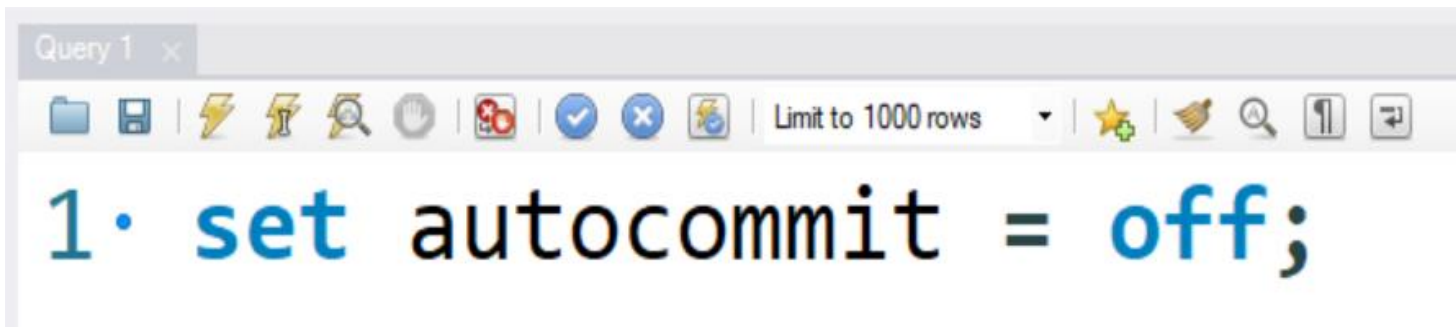
```
1 • rollback;  
2  
3 • select *  
4   from members;
```

Below the query editor, there is a "Result Grid" section. It includes a "Filter Rows:" input field, an "Export:" button, and a "Wrap Cell Content:" checkbox. The result grid displays two rows of data:

	student_number	first_name	last_name	gmail_address	country
▶	20260001	Brenda	Soko	sokobrenda777@gmail.com	Zambia
	20260002	Lemmy	Chanda	chandalemmy01@gmail.com	Zambia

9. ROLLBACK (CONTINUES...)

- By default MySQL is set to AUTOCOMMIT = ON (1), so we need to turn it OFF (0) if you prefer not using the TRANSACTION command.

A screenshot of a MySQL query editor window. The title bar says 'Query 1'. Below the title bar is a toolbar with various icons for file operations, execution, and search. The main text area contains the SQL command '1. set autocommit = off;'. The text is in a monospaced font, with '1.' in black, 'set' in blue, 'autocommit' in black, '=' in black, and 'off;' in blue. The toolbar includes icons for a folder, save, lightning bolt, magnifying glass, hand, redo, undo, checkmark, close, and a dropdown menu showing 'Limit to 1000 rows'. There are also icons for a star, a feather, a search icon, and a refresh icon.

- With this, you don't need explicitly use the TRANSACTION command.

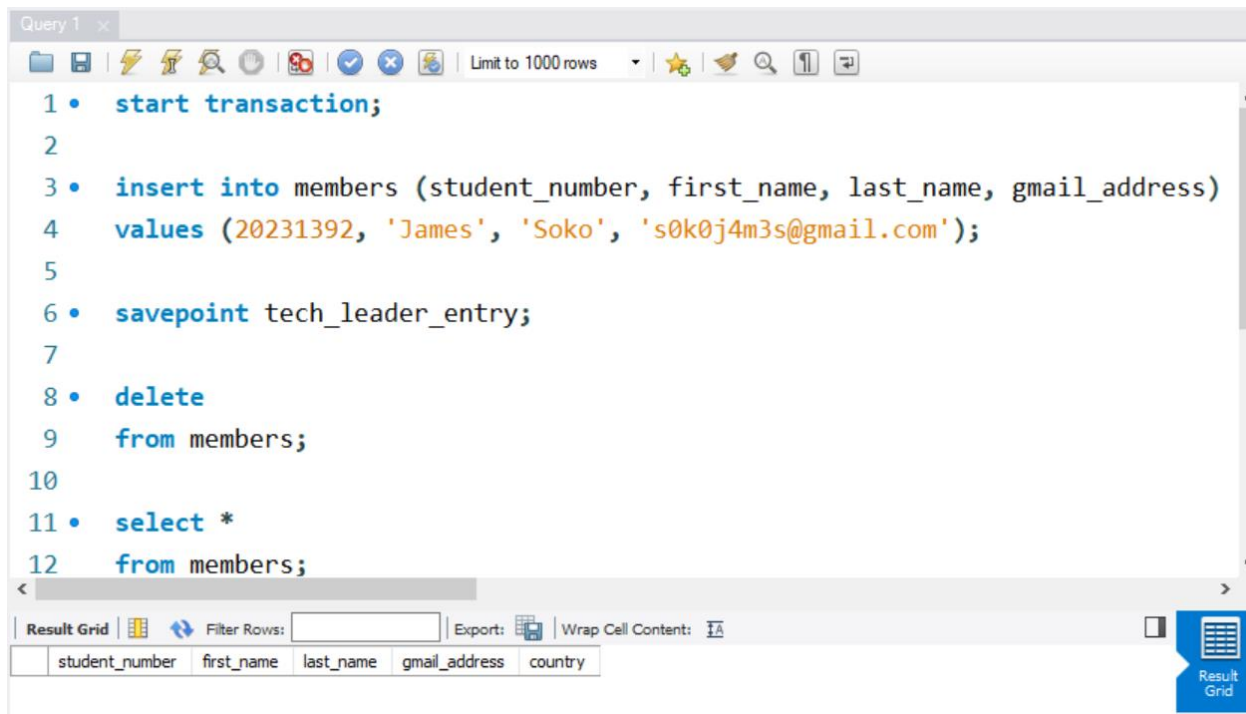
10. SAVEPOINT

- The **SAVEPOINT** statement in MySQL is used to set a point within a transaction to which you can later roll back.

COMMAND	PURPOSE
SAVEPOINT	Creates a named bookmark at the current point in the transaction.

10. SAVEPOINT (CONTINUES...)

- Will deliberately create a SAVEPOINT and delete all records so that we restore the records.



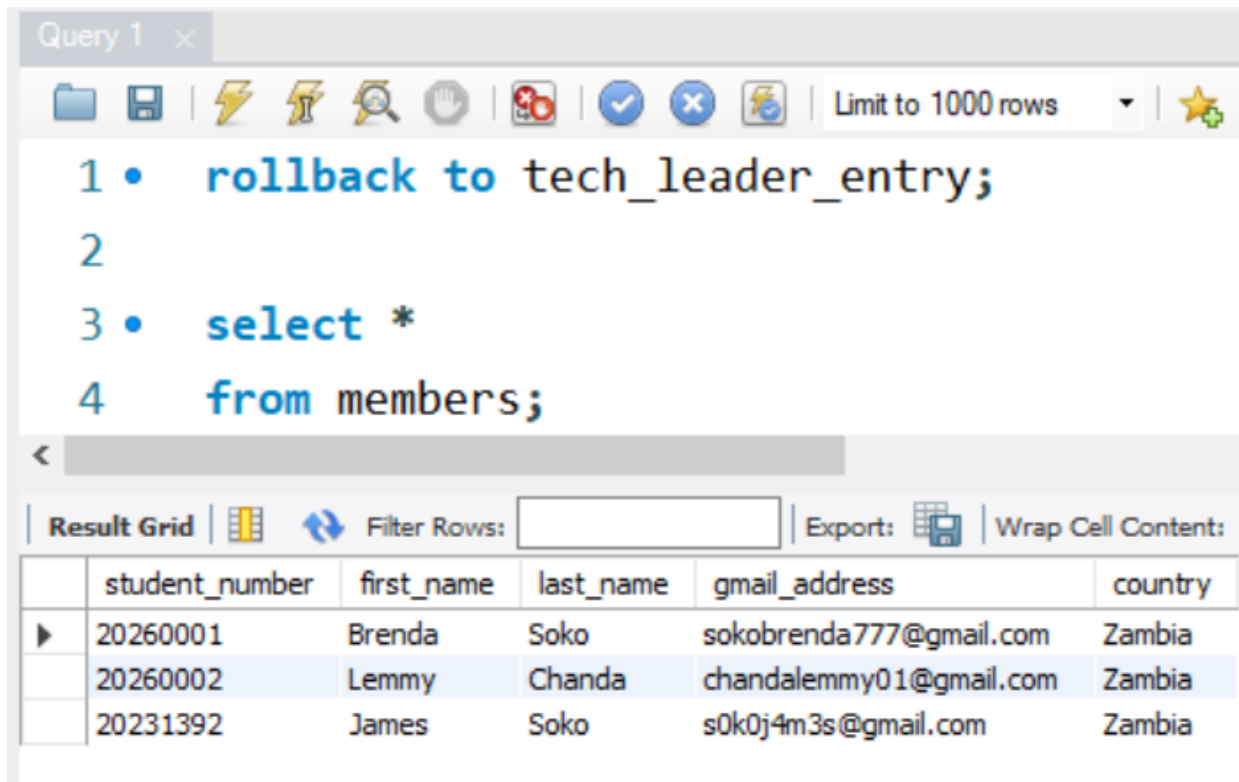
The screenshot shows a SQL query editor window titled "Query 1". The query is as follows:

```
1 • start transaction;
2
3 • insert into members (student_number, first_name, last_name, gmail_address)
4 values (20231392, 'James', 'Soko', 's0k0j4m3s@gmail.com');
5
6 • savepoint tech_leader_entry;
7
8 • delete
9 from members;
10
11 • select *
12 from members;
```

The editor includes a toolbar with various icons and a "Limit to 1000 rows" dropdown. At the bottom, there is a "Result Grid" section with a table header showing columns: student_number, first_name, last_name, gmail_address, and country. A blue "Result Grid" button is also visible in the bottom right corner.

10. SAVEPOINT (CONTINUES...)

- Now we can restore the records –



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

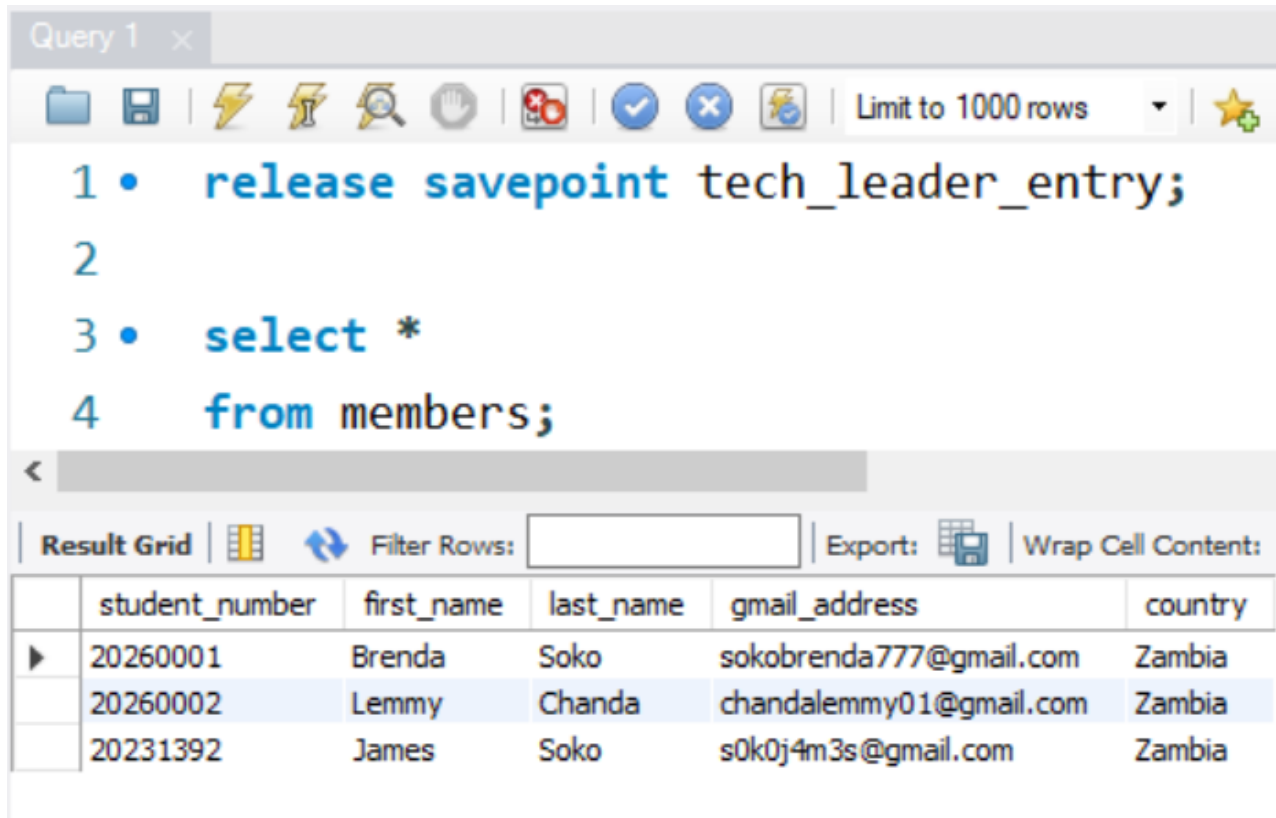
```
1 • rollback to tech_leader_entry;
2
3 • select *
4   from members;
```

Below the query editor is a "Result Grid" showing the results of the query. The grid has six columns: student_number, first_name, last_name, gmail_address, and country. There are three rows of data:

	student_number	first_name	last_name	gmail_address	country
▶	20260001	Brenda	Soko	sokobrenda777@gmail.com	Zambia
	20260002	Lemmy	Chanda	chandalemmmy01@gmail.com	Zambia
	20231392	James	Soko	s0k0j4m3s@gmail.com	Zambia

10. SAVEPOINT (CONTINUES...)

- We can also get rid of the SAVEPOINT by releasing it –



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

```
1 • release savepoint tech_leader_entry;  
2  
3 • select *  
4   from members;
```

Below the query editor, the "Result Grid" is displayed, showing the results of the query. The grid has the following columns: student_number, first_name, last_name, gmail_address, and country. The results are as follows:

	student_number	first_name	last_name	gmail_address	country
▶	20260001	Brenda	Soko	sokobrenda777@gmail.com	Zambia
	20260002	Lemmy	Chanda	chandalemmmy01@gmail.com	Zambia
	20231392	James	Soko	s0k0j4m3s@gmail.com	Zambia

CONCLUSION

- At this point, we now have an understanding about **Transaction Control Language (TCL)** commands.
- In this third handout, our focus was to understand the use of **TRANSACTION, COMMIT, ROLLBACK** and **SAVEPOINT** TCL commands.



UP NEXT... CONSTRAINTS